

group  $g$  and groups that are “not-yet-treated” by time  $t + \delta$ .<sup>7</sup> Importantly, both of these assumptions allow for covariate-specific trends and do not restrict the relationship between treatment timing and the potential outcomes,  $Y_t(g)$ ’s. Thus, they are weaker than the randomization-based assumption made by [Athey and Imbens \(2018\)](#). We also note that the unconditional versions of Assumptions 4 and 5 are weaker than the parallel trends assumption imposed by [de Chaisemartin and D’Haultfoeuille \(2020\)](#) and [Sun and Abraham \(2020\)](#) as they impose fewer restrictions on the evolution of  $Y_t(0)$  in pre-treatment periods; see, e.g., [Marcus and Sant’Anna \(2020\)](#) for a comparison.

In our view, practitioners may favor Assumption 4 with respect to Assumption 5 when there is a sizable group of units that do not participate in the treatment in any period, and, at the same time, these units are similar enough to the “eventually treated” units. When a “never-treated” group of units is not available or “too small”, researchers may favor Assumption 5 as it allows one to use more groups as valid comparison units, which potentially leads to more informative inference procedures. However, it is important to stress that favoring Assumption 5 with respect to Assumption 4 also involves potential drawbacks. For instance, in the absence of treatment anticipation ( $\delta = 0$ ), Assumption 4 does not restrict observed pre-treatment trends across groups, whereas Assumption 5 does; see, e.g., [Marcus and Sant’Anna \(2020\)](#). Not restricting pre-treatment trends may be particularly meaningful in applications where the economic environment during the “early-periods” was potentially different from the “later-periods.” In these cases, the outcomes of different groups may evolve in a non-parallel manner during “early-periods”, perhaps because the groups were exposed to different shocks, while trends become parallel in the “later-periods.” We recommend taking these trade-offs into account when deciding which conditional parallel trends assumption is more appropriate for a given application.<sup>8</sup>

The final identifying assumption we impose is an overlap condition.

**Assumption 6** (Overlap). *For each  $t \in \{2, \dots, T\}$ ,  $g \in \mathcal{G}$ , there exist some  $\varepsilon > 0$  such that  $P(G_g = 1) > \varepsilon$  and  $p_{g,t}(X) < 1 - \varepsilon$  a.s..*

Assumption 6 extends the overlap assumption in [Heckman et al. \(1997, 1998\)](#), [Abadie \(2005\)](#), and [Sant’Anna and Zhao \(2020\)](#) to the multiple groups and multiple periods setup. It states that a positive fraction of the population starts treatment in period  $g$ , and that, for all  $g$  and  $t$ , the generalized propensity score is uniformly bounded away from one. Assumption 6 rules out “irregular identification”, see, e.g., [Khan and Tamer \(2010\)](#).

**Remark 1.** *Note that Assumption 3 and Assumption 4 (Assumption 5) are intrinsically connected. For instance, when one imposes the “no-anticipation” condition (so that  $\delta = 0$ ), Assumption 4 would then impose conditional parallel trends only for post-treatment periods  $t \geq g$ . If one allows for anticipation behavior (so that  $\delta > 0$ ), Assumption 4 would then impose conditional parallel trends in some pre-treatment periods, too. In fact, the parallel trends assumptions become stronger as one increases  $\delta$ . To*

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<sup>7</sup>[Athey and Imbens \(2006\)](#) and [de Chaisemartin and D’Haultfoeuille \(2017\)](#) also consider using “not-yet-treated” units as comparison groups in related DiD procedures.

<sup>8</sup>It may be tempting to use statistical pre-tests to select between different versions of the parallel trends assumption. However, the results of [Roth \(2020\)](#) show that such a practice can lead to important distortions when conducting inference. Thus, we do not recommend following this path, but instead recommend taking the context of the application into account in order to choose the appropriate parallel trends assumption.

the best of our knowledge, this trade-off between the strength of these assumptions has not been noticed before.

**Remark 2.** In some applications, practitioners may not be comfortable with using “never-treated” units as part of the comparison group because they behave very differently from the other “eventually treated” units. In these cases, practitioners could drop all “never-treated” units from the analysis and proceed with Assumption 5.

## 2.4 Nonparametric Identification of the Group-Time Average Treatment Effects

In this section, we show that the family of group-time average treatment effects are nonparametrically point-identified under the aforementioned assumptions. Furthermore, we show that one can use outcome regression (OR), inverse probability weighting (IPW), or doubly robust (DR) estimands to recover the  $ATT(g, t)$ 's. In addition, we also highlight the roles played by Assumption 3 and by Assumptions 4 and 5 when forming these different estimands.

Before formalizing all the results, we need to introduce some additional notation. Let  $m_{g,t,\delta}^{nev}(X) = \mathbb{E}[Y_t - Y_{g-\delta-1} | X, C = 1]$  and  $m_{g,t,\delta}^{ny}(X) = \mathbb{E}[Y_t - Y_{g-\delta-1} | X, D_{t+\delta} = 0, G_g = 0]$ . These are population outcome regressions for the never-treated group and for the “not-yet-treated” by time  $t + \delta$  group. Let

$$ATT_{ipw}^{nev}(g, t; \delta) = \mathbb{E} \left[ \left( \frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_g(X)C}{1-p_g(X)}}{\mathbb{E} \left[ \frac{p_g(X)C}{1-p_g(X)} \right]} \right) (Y_t - Y_{g-\delta-1}) \right], \quad (2.2)$$

$$ATT_{or}^{nev}(g, t; \delta) = \mathbb{E} \left[ \frac{G_g}{\mathbb{E}[G_g]} (Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(X)) \right], \quad (2.3)$$

$$ATT_{dr}^{nev}(g, t; \delta) = \mathbb{E} \left[ \left( \frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_g(X)C}{1-p_g(X)}}{\mathbb{E} \left[ \frac{p_g(X)C}{1-p_g(X)} \right]} \right) (Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(X)) \right]. \quad (2.4)$$

Analogously, let

$$ATT_{ipw}^{ny}(g, t; \delta) = \mathbb{E} \left[ \left( \frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_{g,t+\delta}(X)(1-D_{t+\delta})(1-G_g)}{1-p_{g,t+\delta}(X)}}{\mathbb{E} \left[ \frac{p_{g,t+\delta}(X)(1-D_{t+\delta})(1-G_g)}{1-p_{g,t+\delta}(X)} \right]} \right) (Y_t - Y_{g-\delta-1}) \right], \quad (2.5)$$

$$ATT_{or}^{ny}(g, t; \delta) = \mathbb{E} \left[ \frac{G_g}{\mathbb{E}[G_g]} (Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{ny}(X)) \right], \quad (2.6)$$

$$ATT_{dr}^{ny}(g, t; \delta) = \mathbb{E} \left[ \left( \frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_{g,t+\delta}(X)(1-D_{t+\delta})(1-G_g)}{1-p_{g,t+\delta}(X)}}{\mathbb{E} \left[ \frac{p_{g,t+\delta}(X)(1-D_{t+\delta})(1-G_g)}{1-p_{g,t+\delta}(X)} \right]} \right) (Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{ny}(X)) \right]. \quad (2.7)$$

With some abuse of notation, we write  $\bar{g} - \delta = \infty$  for any non-negative  $\delta$  whenever  $\bar{g} = \infty$ .

**Theorem 1.** *Let Assumptions 1, 2, 3 and 6 hold.*

(i) *If Assumption 4 holds, then, for all  $g$  and  $t$  such that  $g \in \mathcal{G}_\delta$ ,  $t \in \{2, \dots, \mathcal{T} - \delta\}$  and  $t \geq g - \delta$ ,*

$$ATT(g, t) = ATT_{ipw}^{nev}(g, t; \delta) = ATT_{or}^{nev}(g, t; \delta) = ATT_{dr}^{nev}(g, t; \delta).$$

(ii) *If Assumption 5 holds, then, for all  $g$  and  $t$  such that  $g \in \mathcal{G}_\delta$ ,  $t \in \{2, \dots, \mathcal{T} - \delta\}$  and  $g - \delta \leq t < \bar{g} - \delta$ ,*

$$ATT(g, t) = ATT_{ipw}^{ny}(g, t; \delta) = ATT_{or}^{ny}(g, t; \delta) = ATT_{dr}^{ny}(g, t; \delta).$$

Theorem 1 is the first main result of this paper. It provides powerful identification results that extend the DiD identification results based on the outcome regression approach of Heckman et al. (1997, 1998), the IPW approach of Abadie (2005), and the DR approach of Sant’Anna and Zhao (2020) to the multiple-periods, multiple groups setup. In other words, Theorem 1 says that, from an identification point of view, one can recover the  $ATT(g, t)$ ’s by exploiting different parts of the data generating process: the OR approach only relies on modeling the conditional expectation of the outcome evolution for the comparison groups, the IPW approach relies on modeling the conditional probability of being in group  $g$ , whereas the DR approach exploits both OR and IPW components.

In order to extend the results of Heckman et al. (1997, 1998), Abadie (2005), and Sant’Anna and Zhao (2020) to the multiple groups, multiple periods framework, we have to address two different challenges: one associated with an appropriate reference time period and one associated with an appropriate comparison group. Theorem 1 highlights how a solution to these challenges is directly connected to the limited anticipation and the conditional parallel trends assumptions. More specifically, Theorem 1 says that we can use the time period  $t = g - \delta - 1$  as an appropriate reference time period under Assumption 3 and either Assumption 4 or 5. This is the most recent time period when untreated potential outcomes are observed for units in group  $g$ . Interestingly, the more treatment anticipation is allowed (i.e., the higher  $\delta$  is), the further back in time one needs to go.<sup>9</sup> Theorem 1 also suggests that the choice of comparison group is directly tied to the conditional parallel trends assumption one makes: under Assumption 4, one can use “never treated” units as a fixed comparison group for all “eventually treated” units; whereas, under Assumption 5, one can use the “not-yet-treated by time  $t + \delta$ ” units as a valid comparison group for those who are first treated at time  $g$ . In this latter case, Theorem 1 also highlights that when all units eventually gets treated ( $\bar{g} < \infty$ ), one is only able to identify the  $ATT(g, t)$ ’s for time periods before the last treated group “effectively” starts their treatment, i.e.,  $t < \bar{g} - \delta$ . In this case, one can not identify the  $ATT(g, t)$  for the last treated cohort, too.

Finally, we note that when pre-treatment covariates play no role in identification (i.e., Assumptions 3, 4, and 5 hold unconditionally on  $X$ ), (2.2)-(2.4) collapse to

$$ATT_{unc}^{nev}(g, t; \delta) = E[Y_t - Y_{g-\delta-1} | G_g = 1] - E[Y_t - Y_{g-\delta-1} | C = 1], \quad (2.8)$$

and (2.5)-(2.7) collapse to

$$ATT_{unc}^{ny}(g, t; \delta) = E[Y_t - Y_{g-\delta-1} | G_g = 1] - E[Y_t - Y_{g-\delta-1} | D_{t+\delta} = 0]. \quad (2.9)$$

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<sup>9</sup>As mentioned in Remark 1, as one allows  $\delta$  to increase, Assumptions 4 and 5 becomes more restrictive.

These expressions for  $ATT(g, t)$  clearly resemble the one for  $ATT$  in the canonical two-periods and two-groups case. As in that case, the average effect of participating in the treatment for units in group  $g$  is identified by taking the path of outcomes (i.e., the change in outcomes between the most recent period before they were affected by the treatment and the current period) actually experienced by that group and adjusting it by the path of outcomes experienced by a comparison group. Under the parallel trends assumption, this latter path is the path of outcomes that units in group  $g$  would have experienced if they had not participated in the treatment.

**Remark 3.** From (2.8) one can see that when Assumptions 3 and 4 hold unconditionally and there is no-anticipation, the  $ATT(g, t)$  parameter can be obtained by first subsetting the data to only contain observations at time  $t$  and  $g - 1$ , from units with either  $G_g = 1$  or  $C = 1$ , and then, using only the observations of this subset, running the (population) linear regression

$$Y = \alpha_1^{g,t} + \alpha_2^{g,t} \cdot G_g + \alpha_3^{g,t} \cdot 1\{T = t\} + \beta^{g,t} \cdot (G_g \times 1\{T = t\}) + \epsilon^{g,t}. \quad (2.10)$$

It is then easy to verify that  $\beta^{g,t} = ATT(g, t)$ . Note that one would need to consider different partitions of the data to characterize different  $ATT(g, t)$  in terms of regression parameters. Alternatively, one could use the interacted two-way fixed effects regression proposed by [Sun and Abraham \(2020\)](#).

**Remark 4.** When covariates are available, the  $\tilde{\beta}^{g,t}$  coefficient of the population linear regression

$$Y = \tilde{\alpha}_1^{g,t} + \tilde{\alpha}_2^{g,t} \cdot G_g + \tilde{\alpha}_3^{g,t} \cdot 1\{T = t\} + \tilde{\beta}^{g,t} \cdot (G_g \times 1\{T = t\}) + \tilde{\gamma} \cdot X + \tilde{\epsilon}^{g,t}$$

that uses the same subset of data as in Remark 3 is, in general, not equal to  $ATT(g, t)$  unless one is willing to assume (i) homogeneous (in  $X$ ) treatment effects, i.e.,  $\mathbb{E}[Y_t(g) - Y_t(0)|G_g = 1, X] = \mathbb{E}[Y_t(g) - Y_t(0)|G_g = 1]$  a.s., and (ii) rule-out covariate-specific trends, i.e., for  $\mathbb{E}[Y_t - Y_{t-1}|X, G] = \mathbb{E}[Y_t - Y_{t-1}|G]$  a.s. for all groups and time periods; see, e.g., [Śłoczyński \(2018\)](#) for a related discussion. The characterizations of  $ATT(g, t)$  discussed in Theorem 1 do not rely on these restrictive conditions.

**Remark 5.** Although the IPW, OR, and DR based estimands presented in Theorem 1 are identical from an identification standpoint, this is not the case when one wants to estimate and make inference about the  $ATT(g, t)$ . As we discuss in Section 4, DiD estimators based on the DR estimands (2.4) and (2.7) usually enjoy additional robustness against model-misspecifications when compared to the IPW and OR estimands.

**Remark 6.** Theorem 1 suggests that we can identify  $ATT(g, t)$  only for groups in  $\mathcal{G}_\delta \subseteq \mathcal{G}$  which can involve dropping some “early treated” groups due to anticipation effects. When  $\delta = 0$ , i.e. when there is no anticipation,  $\mathcal{G}_\delta = \mathcal{G}$ . Theorem 1 also suggests that we can identify  $ATT(g, t)$  only until  $t = \mathcal{T} - \delta$  because of potential treatment anticipation behavior. In applications where some units are known to never participate in the treatment (including periods after time period  $\mathcal{T}$ ), however, we can identify  $ATT(g, t)$  up to  $t = \mathcal{T}$  by using these units as a valid comparison group for all time periods  $t = \mathcal{T} - \delta + 1, \dots, \mathcal{T}$ , provided that an appropriate parallel trends assumption is satisfied.

**Remark 7.** From Theorem 1 it is clear that pre-treatment covariates play a prominent role in our analysis. Importantly, Assumptions 4 and 5 suggest that researchers should include pre-treatment covariates